

Function Point Analysis Assignment

Project Name: UMass Marketplace

Group Name: Group 10

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Function Point Analysis – Frontend

System Boundary

The application boundary is the **UMass Marketplace frontend interface** developed for users to browse listings, sign in, filter items, view details, message sellers, and create listings. The frontend communicates with backend services and displays marketplace data to users. Since this writeup focuses on the frontend portion, the counted functions are based on the user-visible screens and interactions that I designed and implemented. The overall project and shared system entities are consistent with the team's UMass Marketplace design.

Source of Function Counts

These function point counts were derived from the frontend screens I worked on:

- Welcome page
- Google sign-in page
- Browse marketplace page
- Category filter dropdown
- Price filter dropdown
- Listing detail page
- Safety notice modal
- Messaging screen
- Create new listing screen

1. Internal Logical Files (ILF)

These are logical groups of data used by the frontend and maintained by the overall system database. The frontend interacts with them as core marketplace entities.

ILF Function	Description
Users	User profile and login identity
Listings	Item title, description, price, location, condition, seller, status
Categories	Category options shown in filters and listing creation
Messages / Conversations	Chat history between buyers and sellers
Photos	Listing images shown in browse, detail, and create listing screens

These are considered ILFs because they are user-recognizable data groups that are stored and maintained within the UMass Marketplace system, even though my work specifically focused on the frontend interface that displays and sends data related to them.

ILF Count Table

Type	Count	Complexity	Weight	Total
ILF	5	Average	10	5

2. External Interface Files (EIF)

These are logical data groups used by the system but maintained outside the application boundary.

EIF Function	Description
Google Authentication Service	The frontend includes a “Sign in with Google” flow, which depends on user data maintained by an external authentication provider rather than by the marketplace itself

This is an EIF because the login process references data maintained outside the application boundary of the marketplace frontend/system.

EIF Count Table

Type	Count	Complexity	Weight	Total
EIF	1	Average	7	7

3. External Inputs (EI)

These are user actions where data enters the system and causes a change to internal data or triggers a transaction.

EI Function	Description
Sign in with Google	User initiates login/authentication
Search marketplace	User enters search text
Select category filter	User chooses a category
Select price filter	User chooses a price range
Open message seller flow	User initiates contact from listing page
Acknowledge safety notice	User confirms understanding before proceeding
Send message	User types and sends a chat message

Create listing – enter title	User inputs listing title
Create listing – enter price	User inputs listing price
Create listing – choose category	User selects category
Create listing – enter location	User inputs location
Create listing – choose condition	User selects condition
Create listing – enter description	User inputs description
Upload photos	User uploads listing images
Publish listing	User submits the completed listing

These are External Inputs because they involve data entered by the user through the interface and either update system state, create records, or submit information for processing.

EI Count Table

Type	Count	Complexity	Weight	Total
EI	15	Average	4	60

4. External Outputs (EO)

These are outputs where the frontend presents processed information to the user, usually involving formatting, filtering, or combined display logic.

EO Function	Description
Marketplace browse results	Displays processed listings based on search/filter selections
Filtered listings by category	Shows listings after category-based filtering

Filtered listings by price	Shows listings after price-based filtering
Listing cards on browse page	Displays combined listing info such as image, price, title, and location
Listing detail page	Shows integrated listing information including seller, location, category, description, and images
Messaging conversation display	Shows chat history in a formatted conversation layout
Safety notice modal	Displays a formatted warning and interaction step before messaging
Listing publish confirmation/result	Shows result of submitting a new listing

These are External Outputs because the system sends processed or formatted information to the user, often combining multiple fields or applying filters.

EO Count Table

Type	Count	Complexity	Weight	Total
EO	8	Average	5	40

5. External Inquiries (EQ)

These are retrieval actions where the user requests information and the system returns it without major computation.

EQ Function	Description
View welcome page	Retrieves and displays the initial landing screen
View sign-in page	Retrieves and displays the login interface

Open browse page	Retrieves marketplace listings view
Open category dropdown options	Retrieves available category choices
Open price dropdown options	Retrieves available price filter choices
View specific listing details	Retrieves full information for one listing
View existing messages in a conversation	Retrieves current chat history
Open create listing page	Retrieves the listing creation interface

These are External Inquiries because they primarily retrieve and display information without significant derived processing.

EQ Count Table

Type	Count	Complexity	Weight	Total
EQ	8	Average	4	32

Function Point Calculation Summary

Type	Count	Complexity	Weight	Total
ILF	5	Average	10	50
EIF	1	Average	7	7
EI	15	Average	4	60
EO	8	Average	5	40
EQ	8	Average	4	32

Total Function Points

Unadjusted Function Points (UFP) = 189

Explanation of Complexity

All components were assumed to have **average complexity** because the frontend includes multiple interactive screens, moderate validation, filtered listing views, messaging functionality, listing creation, and integration with an external sign-in provider. The interface is more than a simple display-only UI because it supports both information retrieval and user-driven input workflows.

Summary

This frontend function point analysis reflects the functional size of the user-facing **UMass Marketplace** interface that I worked on. My screens support authentication, browsing, searching, filtering, viewing listing details, messaging sellers, and creating new listings. The analysis aligns with the overall marketplace design used by the team while focusing specifically on the frontend features and flows I implemented.